1. Clinical Question & Core Hypothesis

Clinical Question

The Ontologist highlights three convergent yet discordant biomarker axes in MIBC — TGF-β–driven fibrosis, MSH6-associated mismatch-repair deficiency, and intratumoral immune contexture — raising the question: why do tumors with apparently high neoantigen potential (MMR deficiency) sometimes fail to benefit from immune-mediated therapy when fibrotic features are present? 2) Tier 1 paths used: I built on (a) the TGF-β → fibrogenic/stromal axis because it provides a mechanistic link between signaling and a measurable stromal phenotype, and (b) the MSH6 → MMR deficiency axis because it implicates neoantigen load and potential immunotherapy sensitivity; these together frame a plausible antagonistic interaction. I excluded placing CD4/CD8 TILs inside the core hypothesis to keep the mechanistic claim sharp (they are retained as secondary modifiers). 3) Tier 2 supporting evidence: TMB (links MSH6 to neoantigens and immunogenicity), urine DNA methylation (non-invasive monitoring opportunity), ER/TP53 and metabolic markers (host/tumor modifiers) support evaluating multi-modal stratification and potential confounders of immune response. 4) Endpoint choice rationale: I selected therapy_response (immune-checkpoint/chemo-immunotherapy response) because the central mechanistic question asks why neoantigen-rich tumors sometimes fail to respond — a therapy-outcome problem directly actionable by combining stromal/TGF-β interventions with immunotherapy. 5) Deliberate exclusions: I deliberately excluded CD4/CD8 TILs from the two-gene core to obey the 1–2 anchor limit and keep the hypothesis focused on a stromal vs. genomic tension; I also deferred TMB, urine methylation, ER/TP53, and metabolic markers to secondary_modifiers to be tested as effect modifiers rather than core anchors?

Working Hypothesis

Primary hypothesis: In muscle-invasive bladder cancer (MIBC), a high stromal-to-tumor ratio (TSR), reflecting TGF-β–driven fibrogenesis, establishes a desmoplastic, immunosuppressive microenvironment that prevents neoantigen-rich, MSH6-deficient tumors from mounting effective T cell–mediated anti-tumor responses. Consequently, TSR-high, TGF-β–active, MSH6-deficient tumors will exhibit reduced response to immune-checkpoint therapy (and chemo-immunotherapy) compared with TSR-low or MSH6-intact tumors, despite a high neoantigen burden. This implies that stromal remodeling can override neoantigen-driven immunogenicity. Secondary hypotheses: (i) TGF-β signaling correlates with ECM remodeling and exclusion/functional impairment of CD8+ T cells; (ii) MSH6 deficiency increases neoantigen load but requires adequate T cell infiltration that is impeded by fibrosis; (iii) combining stromal/anti–TGF-β strategies with immunotherapy will restore T cell infiltration and improve clinical response in TSR-high/MSH6-deficient tumors. Operationally, the hypothesis treats CD4/CD8 TIL density as a secondary modifier rather than a core anchor, and tests a multi-layer model where imaging-based TSR and genomic/epigenomic context together predict therapy response. Definitions and measurements: TSR will be quantified from histopathology or radiology images using standardized segmentation (tumor vs stroma ratio); TGF-β pathway activity will be assessed by gene expression signatures and/or phospho-SMAD staining; MSH6 status determined by targeted sequencing and/or IHC; neoantigen load inferred from tumor exome sequencing; therapy_response captured as objective response rate, progression-free survival, and overall survival for immune-checkpoint inhibitor–containing regimens. Testable predictions: (a) In retrospective/prospective cohorts, TSR-high/MSH6-deficient tumors will show lower objective response rates and shorter PFS/OS to ICI therapy than TSR-low or MSH6-intact tumors, even after adjusting for TMB and other modifiers; (b) spatial profiling will reveal reduced CD8+ T cell proximity to tumor cells and elevated TGF-β signaling in TSR-high tumors; (c) preclinical models combining anti–TGF-β/stromal-targeting approaches with ICI will increase T cell infiltration and reduce tumor burden specifically in TSR-high/MSH6-deficient contexts. Clinical translation: validation would justify a biomarker-driven trial of anti–TGF-β/stroma-targeting therapy plus immunotherapy in TSR-high/MSH6-deficient MIBC patients. Overall, the expanded hypothesis integrates a mechanistic two-axis model (stroma/TGF-β vs. genomic neoantigen potential) and a clear path to testing and potential therapeutic refinement.

Element	Value
Imaging feature	Stromal-to-tumor ratio (TSR)
Anchor gene(s)	TGF-β, MSH6
Primary endpoint	therapy_response

2. Why This Gene — Not Others

Primary hypothesis gene(s)

TGF-β: TGF-β–driven fibrosis, MSH6-associated mismatch-repair deficiency, and intratumoral immune contexture — raising the question: why do tumors with apparently high neoantigen potential (MMR deficiency) sometimes fail to benefit from immune-mediated therapy when fibrotic features are present? 2) Tier 1 paths used: I built on (a) the TGF-β → fibrogenic/stromal axis because it provides a mechanistic link between signaling and a measurable stromal phenotype, and (b) the MSH6 → MMR deficiency axis because it implicates neoantigen load and potential immunotherapy sensitivity; these together frame a plausible antagonistic interaction.

MSH6: MSH6-associated mismatch-repair deficiency, and intratumoral immune contexture — raising the question: why do tumors with apparently high neoantigen potential (MMR deficiency) sometimes fail to benefit from immune-mediated therapy when fibrotic features are present? 2) Tier 1 paths used: I built on (a) the TGF-β → fibrogenic/stromal axis because it provides a mechanistic link between signaling and a measurable stromal phenotype, and (b) the MSH6 → MMR deficiency axis because it implicates neoantigen load and potential immunotherapy sensitivity; these together frame a plausible antagonistic interaction.

Alternative / secondary genes considered

Tier	Gene	Role / Reason excluded from core
Secondary	CD4	Supporting modifier from knowledge graph
Secondary	CD8	Supporting modifier from knowledge graph
Secondary	TMB	Supporting modifier from knowledge graph
Secondary	urine DNA methylation biomarkers	Supporting modifier from knowledge graph
Secondary	ER status	Supporting modifier from knowledge graph
Secondary	TP53	Supporting modifier from knowledge graph
Secondary	IGF-1	Supporting modifier from knowledge graph

Secondary	SHBG	Supporting modifier from knowledge graph
Secondary	HDL-C	Supporting modifier from knowledge graph
Secondary	TG	Supporting modifier from knowledge graph
Secondary	ALP	Supporting modifier from knowledge graph

3. Clinically Testable Predictions

Prediction 1: At the tissue level, tumors with the imaging feature but low anchor gene activity will show immune cells confined to structural peripheries rather than infiltrating tumor nests.

4. Minimum Viable Validation Plan

A. Retrospective Cohort Analysis

Data source: TCGA-BLCA (Firehose Legacy), ~400 patients - Cox PH model with interaction term for the imaging feature and anchor gene - Kaplan-Meier 2x2 stratification

B. Tissue-Level Validation

Methods: Multiplex IF / IHC / spatial transcriptomics

Key readouts: - Immune structure markers (e.g., CD20, CD3, PNAd) - Effector cell localization (CD8) relative to tumor nests - Anchor gene protein / phospho-protein expression - Stromal features (collagen density by SHG, if applicable)

Goal: Confirm that the proposed mechanism operates at the spatial / tissue level.

C. Functional Validation

Models: Ex vivo organotypic cultures, 3D co-culture, syngeneic mouse models

Perturbations: Genetic (KO/rescue) or pharmacologic modulation of anchor gene

Readouts: T-cell infiltration, cytotoxicity (granzyme B / IFN-gamma), tumor growth / survival

Goal: Establish causal relationship between anchor gene activity and immune efficacy.

5. Strengths, Caveats & Alternative Directions

Strengths

- Novel mechanistic hypothesis derived from knowledge graph analysis.

Caveats & Risks

- Hypothesis is computationally derived and requires experimental validation.

Alternative Directions

- CD4: Could serve as an alternative mechanistic axis if the primary hypothesis is not supported.
- CD8: Could serve as an alternative mechanistic axis if the primary hypothesis is not supported.